# STACKS, DIFFERENTIAL GRADED SHEAVES: LEMMA-DERIVED-TENSOR-PRODUCT

THE STACKS PROJECT AUTHORS

*Editorial note.* Statement span is lines 4803-4812 of sources/stacks-sdga/pinned-original-source.tex; proof span is lines 4814-4848, the complete author proof (33 body lines).

*Editorial note.* Navigation stubs: 5 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

*Editorial note.* extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

*Editorial note.* Text is retained verbatim, including the author's own typos and placeholders.

*Editorial note.* Queue index 19 of queue_sdga.json (pb=1358, stmt 4803-4812, proof 4814-4848).

*Allowed standard prerequisites (editorial).*

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

*Source reference.* Original lemma 06XN: sources/stacks-sdga/pinned-original-source.tex, source lines 4848–4848. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

*Source reference.* Original lemma 0FTE: sources/stacks-sdga/pinned-original-source.tex, source lines 4795–4800. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

*Source reference.* Original lemma 0FSI: sources/stacks-sdga/pinned-original-source.tex, source lines 3445–3460. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

*Source reference.* Original lemma 0FSC: sources/stacks-sdga/pinned-original-source.tex, source lines 2983–2991. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

*Source reference.* Original lemma 0FSH: sources/stacks-sdga/pinned-original-source.tex, source lines 3186–3196. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

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*Date*: stacks-project:sdga.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

**Lemma 0.1.** *In the situation above, the functor (0FTE) composed with the localization functor $K(Mod(\mathcal{A}', d)) \to D(\mathcal{A}', d)$ has a left derived extension $D(\mathcal{B}, d) \to D(\mathcal{A}', d)$ whose value on a good right differential graded $\mathcal{B}$-module $\mathcal{P}$ is $f^*\mathcal{P} \otimes_{\mathcal{A}} \mathcal{N}$.*

*Proof.* Recall that for any (right) differential graded $\mathcal{B}$-module $\mathcal{M}$ there exists a quasi-isomorphism $\mathcal{P} \to \mathcal{M}$ with $\mathcal{P}$ a good differential graded $\mathcal{B}$-module. See Lemma 0FSH. Hence by Derived Categories, Lemma 06XN it suffices to show that given a quasi-isomorphism $\mathcal{P} \to \mathcal{P}'$ of good differential graded $\mathcal{B}$-modules the induced map

$$f^*\mathcal{P} \otimes_{\mathcal{A}} \mathcal{N} \longrightarrow f^*\mathcal{P}' \otimes_{\mathcal{A}} \mathcal{N}$$

is a quasi-isomorphism. The cone $\mathcal{P}''$ on $\mathcal{P} \to \mathcal{P}'$ is a good differential graded $\mathcal{A}$-module by Lemma 0FSC. Since we have a distinguished triangle

$$\mathcal{P} \to \mathcal{P}' \to \mathcal{P}'' \to \mathcal{P}[1]$$

in $K(Mod(\mathcal{B}, \mathrm{d}))$ we obtain a distinguished triangle

$$f^*\mathcal{P} \otimes_{\mathcal{A}} \mathcal{N} \to f^*\mathcal{P}' \otimes_{\mathcal{A}} \mathcal{N} \to f^*\mathcal{P}'' \otimes_{\mathcal{A}} \mathcal{N} \to f^*\mathcal{P}[1] \otimes_{\mathcal{A}} \mathcal{N}$$

in $K(Mod(\mathcal{A}', \mathrm{d}))$. By Lemma 0FSI the differential graded module $f^*\mathcal{P}'' \otimes_{\mathcal{A}} \mathcal{N}$ is acyclic and the proof is complete. □